

# THE COMPLEXITY OF RECOGNIZING MF GROUPS FROM FINITE PRESENTATIONS

SAUERS

A countable group is MF if it admits finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. We show that the set of finite presentations of MF groups is  $\Pi_2^0$ -complete in the arithmetical hierarchy, and the set of finite presentations of non-MF groups is  $\Sigma_2^0$ -complete; in particular neither set is recursively enumerable. The upper bound is a description of the MF property by finite certificates. The lower bound encodes the set of programs with infinite domain into finite presentations by an effective Higman embedding, a central HNN extension, and a twisted one, using the non-MF group of the companion manuscript on one branch and, on the other, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona.

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CONSUMPTION**

This manuscript is actively being revised and has not been released for public consumption.

## 1. INTRODUCTION

Not every countable group is MF: the companion manuscript *Non-MF Groups* [NMF] constructs a finitely generated simple group  $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$  with property (T) such that every homomorphism from  $H$  to an MF group is trivial. We determine how hard the MF property is to recognize from a finite presentation, in the Kleene–Mostowski arithmetical hierarchy whose levels  $\Sigma_n^0$  and  $\Pi_n^0$  count alternations of quantifiers over a decidable relation.

**Theorem A** (MF recognition is  $\Pi_2^0$ -complete). *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is  $\Pi_2^0$ -complete, and the set of codes of finite presentations of non-MF groups is  $\Sigma_2^0$ -complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable*

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2020 *Mathematics Subject Classification*. Primary 03D40; Secondary 20F10, 46L05, 03D80.

*Key words and phrases*. MF groups, arithmetical hierarchy, finitely presented groups, HNN extensions, Higman embedding, norm matrix coronas.

map  $e \mapsto \hat{R}_e$  into finite presentations such that the group presented by  $\hat{R}_e$  is MF if and only if the  $e$ -th partial computable function has infinite domain.

Recognizing the MF property is therefore exactly as hard as deciding whether a program halts on infinitely many inputs, and remains undecidable relative to an oracle for the halting problem. Figure 1 places the theorem in the arithmetical hierarchy.

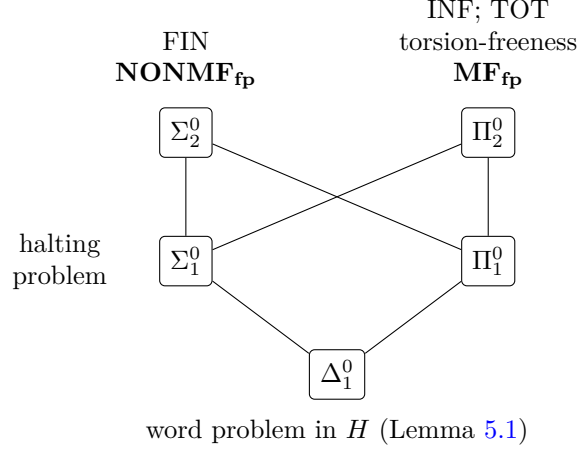


Figure 1: The arithmetical hierarchy. Lines are inclusions, and every listed problem is complete at its level; the word problem in  $H$  is decidable. Boldface marks the two sets of Theorem A: finite certificates put  $\text{MF}_{\text{fp}}$  in  $\Pi_2^0$  (Section 3), and the family  $\hat{R}_e$  reduces  $\text{INF}$  to  $\text{MF}_{\text{fp}}$  and  $\text{FIN}$  to  $\text{NONMF}_{\text{fp}}$  (Sections 5 and 6).

Proposition 3.2 gives a  $\Pi_2^0$  description of the MF presentations: a presentation is MF exactly when finite certificates, each checkable by decidable real arithmetic, exist at every scale (Section 3). For hardness we reduce the  $\Pi_2^0$ -complete set of programs with infinite domain to MF recognition: from  $e$  we construct a finite presentation  $\hat{R}_e$  (Section 5). If the domain is finite, then  $H \leq \hat{R}_e$ , so  $\hat{R}_e$  is not MF. If the domain is infinite, then  $\hat{R}_e$  satisfies the hypotheses of Theorem 4.2, an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona, and is MF (Sections 4 and 6). The MF property is closed under direct limits, as Korchagin proved [Kor, Proposition 13], so every finitely generated group that is not MF is a quotient of a finitely presented one; Theorem A gives the uniform family and the exact level.

**Relation to prior work.** Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property (one that some finitely presented group has and that no group containing some fixed finitely presented group has) can be

recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem A gives the exact level. A classical  $\Pi_2^0$ -complete recognition problem for finitely presented groups is torsion-freeness, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is  $\Pi_2^0$ -hard to recognize [BCR]; their construction is the first step of Section 5. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the HNN extensions of Section 5 are chosen so that Theorem 4.2 applies to them.

## 2. MF GROUPS

A countable group  $G$  is *MF* if there are positive integers  $d_n$  and maps  $V_n: G \rightarrow \mathcal{U}(d_n)$ , with  $V_n(1) = 1$ , such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE]. Here, MF group always means this operator-norm local-approximation property. The strong-convergence convention also requires the matrix models to recover the norms of the left regular representation [GKM, Sch]. Equivalently,  $G$  embeds in the unitary group of the norm matrix corona associated to the sequence  $\mathbf{d} = (d_n)$ ,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}),$$

where  $\bigoplus_n M_{d_n}(\mathbb{C})$  is the  $c_0$ -direct sum of the sequences with  $\|x_n\| \rightarrow 0$ . A separable  $C^*$ -algebra is called MF if it embeds as a  $C^*$ -subalgebra of a norm matrix corona [BK]. Restricting models to a subgroup shows that subgroups of MF groups are MF.

Throughout, the binary Leavitt algebra  $L_{\mathbb{F}_2}(1, 2)$  is the unital  $\mathbb{F}_2$ -algebra generated by  $s_0, s_1, t_0, t_1$  subject to  $t_i s_j = \delta_{ij}$  and  $s_0 t_0 + s_1 t_1 = 1$ , and  $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$  is the subgroup of  $\text{GL}_{12}(L_{\mathbb{F}_2}(1, 2))$  generated by the elementary matrices  $e_{ij}(a) = 1 + aE_{ij}$ ,  $i \neq j$ . By Theorem B of the companion manuscript [NMF], every homomorphism from  $H$  to an MF group is trivial; in particular  $H$  is not MF.

## 3. LOCALITY AND FINITE CERTIFICATES

The MF property is defined through sequences of models on all of  $G$ , but it is determined by finite data with a fixed separation constant. The reformulation is due to Korchagin [Kor].

The following finite-domain formulation is derived from Korchagin [Kor, Proposition 7].

**Lemma 3.1** (locality with fixed separation). *A countable group  $G$  is MF if and only if for every finite set  $F \subseteq G$  containing 1 and every  $\varepsilon > 0$  there are  $d \geq 1$  and a map  $V: F \rightarrow \mathcal{U}(d)$  with  $V(1) = 1$  such that*

$$\begin{aligned} \|V(gh) - V(g)V(h)\| &\leq \varepsilon \quad \text{whenever } g, h, gh \in F, \\ \|V(g) - 1\| &\geq 1 \quad \text{for } g \in F \setminus \{1\}. \end{aligned}$$

*Proof.* Maps for an exhaustion  $(F_n, 1/n)$  of  $G$ , extended by 1 outside  $F_n$ , form an MF model. Conversely, Korchagin’s amplification — replacing  $V_n(g)$  by a direct sum of its tensor powers — raises the separation of finitely many nonidentity elements above 1 while multiplying the defects by a constant [Kor, Proposition 7]. A block direct sum of late amplified models, one summand for each  $g \in F \setminus \{1\}$ , gives the stated map.  $\square$

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set  $A \subseteq \mathbb{N}$  is  $\Sigma_1^0$  if  $A = \{x : \exists y R(x, y)\}$  for a decidable relation  $R$ , it is  $\Pi_1^0$  if its complement is  $\Sigma_1^0$ , and  $\Sigma_2^0$  and  $\Pi_2^0$  are the classes  $\{x : \exists y \forall z R(x, y, z)\}$  and  $\{x : \forall y \exists z R(x, y, z)\}$  with  $R$  decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write  $G_P$  for the group presented by the code  $P$ . Let  $\text{MF}_{\text{fp}}$  be the set of codes  $P$  such that  $G_P$  is MF, and  $\text{NONMF}_{\text{fp}}$  its complement. Korchagin’s Proposition 8 [Kor] is the finite-presentation form of Lemma 3.1; the certificates below add normal-closure witnesses for the trivial words, which is what makes the relation decidable.

**Proposition 3.2** (finite certificates). *There is a decidable relation  $C(P, n, c)$  on triples of natural numbers such that a code  $P$  lies in  $\text{MF}_{\text{fp}}$  if and only if for every  $n$  there is  $c$  with  $C(P, n, c)$ . Consequently  $\text{MF}_{\text{fp}} \in \Pi_2^0$  and  $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ .*

*Proof.* Let  $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m \rangle$ . A *certificate at scale  $n$*  is a triple  $c = (d, \ell, \pi)$  consisting of a dimension  $d \geq 1$ , a labelling  $\ell$  of every reduced word of length at most  $n$  in  $x_1^{\pm 1}, \dots, x_k^{\pm 1}$  by one of the symbols **T** and **S**, and, for every word  $w$  labelled **T**, an expression of  $w$  as a product of conjugates of the relators and their inverses that freely reduces to  $w$ . Such an expression is checkable, and its existence forces  $w = 1$  in  $G_P$ . Let  $\Phi(P, n, c)$  be the statement that there are  $U_1, \dots, U_k \in \mathcal{U}(d)$  with  $\|r_i(U) - 1\| \leq 2^{-n}$  for  $i \leq m$  and  $\|w(U) - 1\| \geq 1/4$  for every word  $w$  labelled **S**, where  $w(U)$  is the evaluation of  $w$  at  $U_1, \dots, U_k$  with  $x_j^{-1} \mapsto U_j^*$ . Writing  $U_j = A_j + iB_j$  with real matrices, unitarity is a system of polynomial equations, and a bound  $\|X\| \leq t$  or  $\|X\| \geq t$  on a matrix whose entries are polynomials in the variables is a semialgebraic condition. Indeed,  $\|X\| \leq t$  holds precisely when  $t^2 - X^*X$  is positive semidefinite, while, for  $t \geq 0$ ,  $\|X\| \geq t$  holds precisely when there is a vector  $v$  with  $\|v\| = 1$  and  $\|Xv\|^2 \geq t^2$ . Thus  $\Phi(P, n, c)$  is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski’s theorem [Ta]. Let  $C(P, n, c)$  hold when  $c$  is a

well-formed certificate at scale  $n$  whose  $\mathsf{T}$ -expressions check and  $\Phi(P, n, c)$  holds. Then  $C$  is decidable.

Suppose that  $G_P$  is MF. Fix  $n$ , let  $L = \max(\{1, n\} \cup \{|r_i| : 1 \leq i \leq m\})$ , let  $F$  be the set of images in  $G_P$  of all words of length at most  $L$ , together with their inverses, and let  $\varepsilon = 2^{-n}/(4L)$ . Lemma 3.1 gives  $V: F \rightarrow \mathcal{U}(d)$  with  $V(1) = 1$ , defect at most  $\varepsilon$  on  $F$ , and  $\|V(g) - 1\| \geq 1$  for  $g \in F \setminus \{1\}$ . Put  $U_j = V(\bar{x}_j)$ , where  $\bar{w}$  denotes the image of a word  $w$  in  $G_P$ . For  $g \in F$  we have  $g^{-1} \in F$  and  $\|V(g)V(g^{-1}) - 1\| \leq \varepsilon$ , so  $\|V(g^{-1}) - V(g)^*\| \leq \varepsilon$ . For a word  $w = y_1 \cdots y_\ell$  of length  $\ell \leq L$ , all of whose prefixes have images in  $F$ , induction on  $\ell$  gives  $\|w(U) - V(\bar{w})\| \leq 2\ell\varepsilon$ . In particular  $\|r_i(U) - 1\| \leq 2L\varepsilon \leq 2^{-n}$ . Label a word  $w$  of length at most  $n$  by  $\mathsf{T}$  if  $\bar{w} = 1$ , with a normal-closure expression as witness, and by  $\mathsf{S}$  otherwise; for the latter,  $\|w(U) - 1\| \geq \|V(\bar{w}) - 1\| - 2n\varepsilon \geq 1 - \frac{1}{2} \geq \frac{1}{4}$ . So  $C(P, n, c)$  holds for this certificate.

Conversely, suppose that for every  $n$  some certificate  $c_n = (d_n, \ell_n, \pi_n)$  satisfies  $C(P, n, c_n)$ , and choose  $U^{(n)} \in \mathcal{U}(d_n)^k$  witnessing  $\Phi(P, n, c_n)$ . For each  $g \in G_P$  fix a reduced word  $w_g$  representing  $g$ , with  $w_1$  the empty word, and put  $V_n(g) = w_g(U^{(n)})$ . Then  $V_n(1) = 1$ . For  $g, h \in G_P$  the word  $t = w_g w_h w_{gh}^{-1}$  is trivial in  $G_P$ , so it is freely equal to a product of  $A$  conjugates of relators and their inverses, for some  $A$  depending on  $g$  and  $h$  but not on  $n$ . Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since  $\|xy - 1\| \leq \|x - 1\| + \|y - 1\|$  for unitaries  $x, y$ ,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \|t(U^{(n)}) - 1\| \leq A 2^{-n} \rightarrow 0.$$

For  $g \neq 1$  the word  $w_g$  is nontrivial in  $G_P$ , so for  $n \geq |w_g|$  the certificate  $c_n$  cannot label it  $\mathsf{T}$ , hence labels it  $\mathsf{S}$ , and  $\|V_n(g) - 1\| \geq 1/4$ . Thus  $\limsup_n \|V_n(g) - 1\| \geq 1/4 > 0$ , and  $G_P$  is MF.

The displayed equivalence exhibits  $\text{MF}_{\text{fp}}$  as  $\{P : \forall n \exists c C(P, n, c)\}$ , a  $\Pi_2^0$  set, and its complement as a  $\Sigma_2^0$  set.  $\square$

#### 4. HNN EXTENSIONS WITH A CORONA CONJUGATOR

In the companion manuscript [NMF], we prove non-approximability using one-sided conjugation in a norm matrix corona. If the edge isomorphism of an HNN extension is implemented by a unitary of a norm matrix corona, then the HNN extension is MF. For  $\theta = \text{id}$  this is close to Shulman's theorem on central HNN extensions [Sh]; the version below assumes a tracial state vanishing off the identity and allows a nontrivial edge isomorphism. The corona unitary implements the edge isomorphism, while the trace and the left regular representation prove that the group embeds in the resulting algebra.

A countable group  $G$  admits a *tracial MF realization*  $(A, \rho, \tau)$  if  $A$  is a separable unital MF  $C^*$ -algebra,  $\rho: G \rightarrow \mathcal{U}(A)$  is a homomorphism, and  $\tau$  is a tracial state on  $A$  with  $\tau(\rho(g)) = 0$  for every  $g \neq 1$ . Since  $\tau(1) = 1$ , such a group embeds in  $\mathcal{U}(A)$ , so it is MF. Only the vanishing of  $\tau$  off the identity

is used below; traciality is what the constructions of Section 6 provide. To embed an arbitrary norm matrix corona  $\mathcal{Q}_{\mathbf{d}}$  in  $\prod_k M_k(\mathbb{C})/\bigoplus_k M_k(\mathbb{C})$ , choose a strictly increasing sequence  $(m_n)$  with  $m_n \geq d_n$ , embed  $M_{d_n}(\mathbb{C})$  as a corner of  $M_{m_n}(\mathbb{C})$ , and put zero in the unused coordinates. The resulting map on reduced products is an isometric  $*$ -homomorphism, so the MF algebras of Section 2 are exactly the separable  $C^*$ -algebras that embed in the latter corona, as in [BK, Sh].

**Lemma 4.1** (residually finite groups). *Every countable residually finite group admits a tracial MF realization.*

*Proof.* Let  $G$  be countable and residually finite, and choose finite-index normal subgroups  $N_1 \geq N_2 \geq \dots$  with  $\bigcap_k N_k = \{1\}$ ; this is possible because  $G$  is countable. Let  $d_k = [G : N_k]$  and let  $\lambda_k: G \rightarrow \mathcal{U}(d_k)$  be the left regular representation of  $G/N_k$ . In the corona  $\mathcal{Q}_{\mathbf{d}}$  put  $\rho(g) = [(\lambda_k(g))_k]$ ; then  $\rho$  is a homomorphism into the unitary group. Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  and set  $\tau([(x_k)_k]) = \lim_{\omega} \text{tr}_{d_k}(x_k)$ , where  $\text{tr}_{d_k}$  is the normalized trace on  $M_{d_k}(\mathbb{C})$ ; this is well defined because  $|\text{tr}_{d_k}(x_k)| \leq \|x_k\|$ , and it is a tracial state on  $\mathcal{Q}_{\mathbf{d}}$ . If  $g \neq 1$ , then  $g \notin N_k$  for all large  $k$ , so  $\text{tr}_{d_k}(\lambda_k(g)) = 0$  for all large  $k$  and  $\tau(\rho(g)) = 0$ . Let  $A$  be the  $C^*$ -algebra generated by  $\rho(G)$  and the unit of  $\mathcal{Q}_{\mathbf{d}}$ ; it is separable, unital, and MF, and  $(A, \rho, \tau|_A)$  is a tracial MF realization of  $G$ .  $\square$

Let  $G$  be a countable group, let  $S \leq G$  be a subgroup, and let  $\theta: S \rightarrow G$  be an injective homomorphism. The HNN extension

$$(1) \quad R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S) \rangle$$

contains  $G$  by Britton's lemma [Bri].

**Theorem 4.2** (HNN permanence with a corona conjugator). *In the situation of (1), suppose that  $G$  admits a tracial MF realization  $(A, \rho, \tau)$ , and that for some norm matrix corona  $\mathcal{Q}$  there are an injective  $*$ -homomorphism  $\iota: A \rightarrow \mathcal{Q}$  and a unitary  $W \in \mathcal{Q}$  with*

$$W \iota \rho(s) W^* = \iota \rho(\theta(s)) \quad (s \in S).$$

*Then  $R$  admits a tracial MF realization. In particular,  $R$  is MF.*

*Proof.* Write  $D = C^*(\iota \rho(G)) \subseteq \mathcal{Q}$ , a separable  $C^*$ -algebra with unit  $p = \iota(1)$ , and let  $B_0 = C^*(\iota \rho(S))$  and  $B_1 = C^*(\iota \rho(\theta(S)))$ , both containing  $p$ . Conjugation by  $W$  restricts to a  $*$ -isomorphism  $\Theta: B_0 \rightarrow B_1$  carrying  $\iota \rho(s)$  to  $\iota \rho(\theta(s))$ . Let  $U$  be the quotient of the full unital free product  $D * C(\mathbb{T})$  by the closed ideal generated by the elements  $ubu^* - \Theta(b)$ ,  $b \in B_0$ , where  $u$  is the canonical unitary generator of  $C(\mathbb{T})$ . We write  $d$  and  $u$  for the images of  $d \in D$  and  $u$  in  $U$ . By construction,  $U$  has the following universal property: whenever  $\pi: D \rightarrow E$  is a unital  $*$ -homomorphism into a unital  $C^*$ -algebra and  $v \in \mathcal{U}(E)$  satisfies  $v\pi(b)v^* = \pi(\Theta(b))$  for  $b \in B_0$ , there is a unique  $*$ -homomorphism  $U \rightarrow E$  extending  $\pi$  and sending  $u$  to  $v$ . We first embed  $U$

in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of  $R$  inside  $U$ .

*Step 1:  $U$  embeds in an amalgamated free product.* This step follows a corner argument of Ueda [Ue]. Let  $C = B_0 \oplus B_1$ ,  $A_1 = M_2(D)$ , and  $A_2 = M_2(B_0)$ , with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \quad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let  $P = A_1 *_C A_2$  be the full amalgamated free product, in which  $A_1$  and  $A_2$  embed [Sh, Section 2.1]. Write  $e_{ij}$  and  $f_{ij}$  for the matrix units of  $A_1$  and  $A_2$ . Since  $\iota_A(p, 0) = \iota_B(p, 0)$  and  $\iota_A(0, p) = \iota_B(0, p)$ , we have  $e_{11} = f_{11} =: e$  and  $e_{22} = f_{22}$  in  $P$ . Define  $\pi_D: D \rightarrow ePe$  by  $\pi_D(d) = \text{diag}(d, 0)$ , a unital  $*$ -homomorphism into the corner, and put  $w = e_{12}f_{21} \in ePe$ . Then  $ww^* = e_{12}f_{22}e_{21} = e_{12}e_{22}e_{21} = e$  and  $w^*w = f_{12}e_{22}f_{21} = f_{12}f_{22}f_{21} = e$ , so  $w$  is a unitary of  $ePe$ . For  $b \in B_0$  the element  $\text{diag}(b, 0)$  is  $\iota_A(b, 0) = \iota_B(b, 0)$ , so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}(f_{21} \text{diag}(b, 0)f_{12})e_{21} = e_{12} \iota_B(0, \Theta(b)) e_{21} \\ &= e_{12} \iota_A(0, \Theta(b)) e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses  $\iota_B(0, \Theta(b)) = \text{diag}(0, b)$  in  $A_2$ . The universal property of  $U$  gives a  $*$ -homomorphism  $\Phi: U \rightarrow ePe \subseteq P$  with  $\Phi|_D = \pi_D$  and  $\Phi(u) = w$ . We prove that  $\Phi$  is injective by constructing  $\Psi: P \rightarrow M_2(U)$  with  $\Psi\Phi(x) = \text{diag}(x, 0)$ . Let  $\Psi_1: A_1 \rightarrow M_2(U)$  be the entrywise inclusion  $D \subseteq U$ , and let  $\Psi_2: A_2 \rightarrow M_2(U)$  be the entrywise inclusion  $B_0 \subseteq U$  followed by conjugation by  $\text{diag}(1, u)$ . On  $C$  we have  $\Psi_1(\iota_A(c_0, c_1)) = \text{diag}(c_0, c_1)$  and  $\Psi_2(\iota_B(c_0, c_1)) = \text{diag}(c_0, u\Theta^{-1}(c_1)u^*) = \text{diag}(c_0, c_1)$ , so the pair induces a  $*$ -homomorphism  $\Psi: P \rightarrow M_2(U)$ . On generators,  $\Psi\Phi(d) = \text{diag}(d, 0)$  and  $\Psi\Phi(u) = \Psi_1(e_{12})\Psi_2(f_{21}) = \text{diag}(u, 0)$ . Hence  $\Psi\Phi$  is the injective  $*$ -homomorphism  $x \mapsto \text{diag}(x, 0)$  of  $U$  into  $M_2(U)$ , and  $\Phi$  is injective.

*Step 2:  $P$  is MF.* By Shulman's amalgamated-free-product criterion [Sh, Theorem 16], the full amalgamated free product  $A_1 *_C A_2$  of separable  $C^*$ -algebras is MF as soon as there are injective  $*$ -homomorphisms  $\varphi_A$  and  $\varphi_B$  of  $A_1$  and  $A_2$  into one norm matrix corona with  $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$ . Identify  $M_2(\mathcal{Q})$  with the norm matrix corona of the doubled dimension sequence, and let  $\varphi_A: M_2(D) \rightarrow M_2(\mathcal{Q})$  be the entrywise inclusion and  $\varphi_B: M_2(B_0) \rightarrow M_2(\mathcal{Q})$  the entrywise inclusion followed by conjugation by  $\text{diag}(1, W)$ . Both are injective, and on  $C$ ,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because  $\Theta$  is conjugation by  $W$ . Hence  $P$  is MF, and so are its  $C^*$ -subalgebras  $ePe$  and  $\Phi(U) \cong U$ .

*Step 3: the tracial state.* The functional  $T = \tau \circ \iota^{-1}$  is a tracial state on  $\iota(A) \supseteq D$  with  $T(\iota\rho(g)) = 0$  for  $g \neq 1$ . In the GNS representation of  $D$  for  $T$ , the vectors  $\xi_g = \iota\rho(g)\xi_T$ ,  $g \in G$ , are orthonormal, because  $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$ . Their closed span is invariant under  $D$ , and on it



$\iota\rho(g)$  acts as the left regular representation  $\lambda_G(g)$  of  $G$ . Thus there is a  $*$ -homomorphism  $\pi: D \rightarrow C_r^*(G)$  with  $\pi(\iota\rho(g)) = \lambda_G(g)$ . Since  $G \leq R$ , the space  $\ell^2(R)$  is a direct sum of copies of  $\ell^2(G)$  indexed by the right cosets of  $G$ , so  $\lambda_R|_G$  is a multiple of  $\lambda_G$  and the assignment  $\lambda_G(g) \mapsto \lambda_R(g)$  extends to a  $*$ -homomorphism  $C_r^*(G) \rightarrow C_r^*(R)$ . Composing, we obtain  $\sigma_0: D \rightarrow C_r^*(R)$  with  $\sigma_0(\iota\rho(g)) = \lambda_R(g)$ . The pair  $(\sigma_0, \lambda_R(t))$  satisfies the covariance relation  $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$  for  $b \in B_0$ : on the generators  $\iota\rho(s)$  both sides equal  $\lambda_R(\theta(s))$ , because  $tst^{-1} = \theta(s)$  in  $R$ , and two  $*$ -homomorphisms that agree on generators agree on  $B_0$ . The universal property of  $U$  gives  $\sigma: U \rightarrow C_r^*(R)$  with  $\sigma(\iota\rho(g)) = \lambda_R(g)$  and  $\sigma(u) = \lambda_R(t)$ .

Define  $j: R \rightarrow \mathcal{U}(U)$  by  $j(g) = \iota\rho(g)$  for  $g \in G$  and  $j(t) = u$ . The defining relations of (1) hold in  $U$ , since  $u\iota\rho(s)u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$ , so  $j$  is a homomorphism, and  $\sigma \circ j = \lambda_R$  is injective, so  $j$  is injective. Let  $A' = C^*(j(R)) \subseteq U$ , a separable unital MF  $C^*$ -algebra, and let  $\tau' = \tau_R \circ \sigma|_{A'}$ , where  $\tau_R$  is the canonical trace  $x \mapsto \langle x\delta_1, \delta_1 \rangle$  of  $C_r^*(R)$ . Then  $\tau'$  is a tracial state on  $A'$  with  $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$  for  $r \neq 1$ . Hence  $(A', j, \tau')$  is a tracial MF realization of  $R$ .  $\square$

Injectivity of  $j$  follows from  $\sigma \circ j = \lambda_R$ , not from the covariant pair  $(\iota\rho, W)$ , which need not be faithful; in the applications below it is not.

**Corollary 4.3** (central HNN extensions). *If  $G$  admits a tracial MF realization and  $S \leq G$  is any subgroup, then  $\langle G, t \mid [t, s] = 1 \ (s \in S) \rangle$  admits a tracial MF realization.*

*Proof.* Apply Theorem 4.2 with  $\theta$  the inclusion of  $S$ , with  $\iota$  any injective  $*$ -homomorphism of  $A$  into a norm matrix corona, and with  $W = 1$ .  $\square$

## 5. THE COMPLEXITY OF RECOGNIZING MF GROUPS

Write  $W_e$  for the domain of the  $e$ -th partial computable function and

$$\text{INF} = \{e : W_e \text{ is infinite}\}, \quad \text{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is  $\Pi_2^0$ -complete and FIN is  $\Sigma_2^0$ -complete [So, Chapter IV]. The construction combines the non-MF group  $H$  of the companion manuscript [NMF] with Theorem 4.2: it produces a computable map  $e \mapsto \widehat{R}_e$  into finite presentations such that the presented group contains  $H$  when  $e \in \text{FIN}$  and satisfies the hypotheses of Theorem 4.2 when  $e \in \text{INF}$ , so that it is MF if and only if  $e \in \text{INF}$ . The construction of Bilanovic–Chubb–Roven gives a recursively presented group  $C_e$  that is trivial when  $e \in \text{INF}$  and contains  $H$  when  $e \in \text{FIN}$ . We embed  $C_e$  in a three-generated recursively presented group  $Q_e = F/N_e$ , use a Mihailova subgroup to detect  $N_e$  inside a finitely presented group, and apply two HNN extensions, in the manner of Higman’s rope trick [Hig], to obtain  $\widehat{R}_e$ . For  $e \in \text{FIN}$ , the group  $\widehat{R}_e$  is not MF; for  $e \in \text{INF}$ , Section 6 proves that it is MF. Table 1 lists the objects of the construction.



|                               |                                                                                                                                                           |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| $H$                           | the non-MF group $\text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$ of [NMF]                                                                                        |
| $C_e$                         | recursively presented; trivial for $e \in \text{INF}$ , contains $H$ for $e \in \text{FIN}$                                                               |
| $Q_e = F/N_e$                 | three-generated recursively presented group containing $C_e$ ; here $F = F(x, y, t)$                                                                      |
| $Q_+ = F/N_+$                 | the value of $Q_e$ for trivial $C_e$ ; embeds in $F(x_1, y) \times F(x_2, t)$                                                                             |
| $\Lambda_e; M_e$              | finitely presented Higman host of $Q_e$ ; the Mihailova subgroup detecting equality in $\Lambda_e$                                                        |
| $K_e; L_e$                    | a product of six free groups; its finitely generated subgroup with $i(F) \cap L_e = i(N_e)$                                                               |
| $\Gamma_e; S_e$               | the central HNN extension $\langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle$ ;<br>$S_e = \langle i(F), v i(F) v^{-1} \rangle \cong F *_{N_e} F$ |
| $\tau_e: S_e \rightarrow Q_e$ | $q_e$ on $i(F)$ , trivial on $v i(F) v^{-1}$                                                                                                              |
| $\hat{R}_e$                   | the finite presentation of $\langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \tau_e(s)) \ (s \in S_e) \rangle$                                     |

Table 1: The objects of the construction, in order of appearance.

### A recursive presentation of $H$ .

**Lemma 5.1** (a recursive presentation of  $H$ ). *The group  $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$  is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

*Proof.* Let  $Y$  be the set of elements  $e_{ij}(a)$  with  $i \neq j$  and  $a \in \{1, s_0, s_1, t_0, t_1\}$ . Since  $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$  and  $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$  for distinct  $i, j, k$ , induction on monomial length shows that  $H = \langle Y \rangle$ . The monomials in  $s_0, s_1, t_0, t_1$  that contain no subword  $t_i s_j$  and no subword  $s_1 t_1$  form the standard reduced  $\mathbb{F}_2$ -basis of  $L_{\mathbb{F}_2}(1, 2)$  [Lev]. Indeed, orient the defining relations as  $t_i s_j \rightarrow \delta_{ij}$  and  $s_1 t_1 \rightarrow 1 + s_0 t_0$ , using a degree-lexicographic order in which  $s_1 > s_0$ . The critical words are  $t_i s_1 t_1$  and  $s_1 t_1 s_j$ . Their two reductions agree because, for  $i, j \in \{0, 1\}$ ,

$$\delta_{i1} t_1 = t_i + \delta_{i0} t_0, \quad \delta_{1j} s_1 = s_j + \delta_{0j} s_0.$$

The Diamond Lemma [Ber] gives precisely the displayed irreducible monomials as a basis. Products of reduced monomials are computed by the defining reductions, so equality in  $L_{\mathbb{F}_2}(1, 2)$  is decidable. A word in  $Y^{\pm 1}$  is therefore evaluated to an explicit  $12 \times 12$  matrix over  $L_{\mathbb{F}_2}(1, 2)$ , and it represents 1 in  $H$  exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and  $\langle Y \mid \text{that set} \rangle$  is a recursive presentation of  $H$ .  $\square$

### The group $C_e$ .

**Lemma 5.2** (the group  $C_e$ ). *There is an algorithm that computes from  $e$  a recursive presentation, on a computable set of generators, of a group  $C_e$  such that  $C_e$  is trivial if  $e \in \text{INF}$  and  $H$  embeds in  $C_e$  if  $e \in \text{FIN}$ .*

*Proof.* Apply the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] to the trivial group and  $H$ . Let  $\langle Y \mid T \rangle$  be the recursive presentation of  $H$  from Lemma 5.1. Take generators  $y_{i,\ell}$  for  $i \geq 1$  and  $\ell \in Y$ , one copy  $Y_i$  of  $Y$  for each  $i$ , and as relators all words of  $T$  rewritten in each copy  $Y_i$ , together with every generator  $y_{i,\ell}$  for which  $i \leq |W_e|$ . The last family is recursively enumerable uniformly in  $e$ : enumerate  $W_e$ , and whenever its  $i$ -th element appears, enumerate all of  $Y_i$ . The group presented is the free product of the copies  $H_i \cong H$ ,  $i \geq 1$ , with the first  $|W_e|$  copies killed. If  $W_e$  is infinite every copy is killed and  $C_e$  is trivial; if  $W_e$  is finite,  $C_e$  is the free product of the remaining copies and contains  $H$ .  $\square$

**A three-generator embedding.** Let  $F = F(x, y, t)$  be the free group on  $x, y, t$ , and let  $y_i = x^i y x^{-i}$  for  $i \in \mathbb{Z}$ . For a countable group  $C$  with a generating sequence  $(c_i)_{i \geq 1}$ , put  $c_i = 1$  for  $i \leq 0$  and

$$(2) \quad B(C) = \langle C * F(x, y), t \mid t y_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle.$$

Let  $Q_+ = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$ , let  $q_+ : F \rightarrow Q_+$  be the quotient map, and let  $N_+ = \ker q_+$ .

**Lemma 5.3** (the three-generator embedding).

- (1)  $B(C)$  is an HNN extension of  $C * F(x, y)$ , so  $C$  embeds in  $B(C)$ ; it is generated by  $x, y, t$ ; and a recursive presentation of  $B(C)$  on  $x, y, t$  is computable from a recursive presentation of  $C$ . We write  $B(C) = F/N_C$  and  $q_C : F \rightarrow B(C)$  for the quotient map.
- (2) Killing  $C$  induces a surjection  $B(C) \rightarrow Q_+$ ; hence  $N_C \leq N_+$ , with equality when  $C$  is trivial.
- (3) The assignment  $x \mapsto (x_1, x_2)$ ,  $y \mapsto (y, 1)$ ,  $t \mapsto (1, t)$  defines an injective homomorphism  $j : Q_+ \rightarrow P = F(x_1, y) \times F(x_2, t)$ . In particular,  $Q_+$  is residually finite.

*Proof.* (1) The elements  $y_i$ ,  $i \in \mathbb{Z}$ , freely generate the normal closure of  $y$  in  $F(x, y)$  [LS, Chapter I]. The retraction  $C * F(x, y) \rightarrow F(x, y)$  killing  $C$  sends  $c_i y_i$  to  $y_i$ , so a nontrivial reduced word in the  $c_i y_i$  maps to a nontrivial word in the  $y_i$ ; hence the  $c_i y_i$  freely generate a free subgroup, and  $y_i \mapsto c_i y_i$  is an isomorphism between two free subgroups of the base. Thus (2) is an HNN extension, and the base, hence  $C$ , embeds by Britton’s lemma. The relation  $c_i = t y_i t^{-1} y_i^{-1}$  shows that  $x, y, t$  generate  $B(C)$ , and substituting this expression for  $c_i$  into the relators of  $C$  and into (2) produces a recursive presentation on  $x, y, t$ , uniformly in the presentation of  $C$ .

(2) The relators of (2) map to the relators  $[t, y_i]$  of  $Q_+$  when every  $c_i$  is sent to 1, and the map is the identity on  $x, y, t$ .

(3) The relators  $[t, y_i]$  are sent to  $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$ , so  $j$  is a homomorphism. Let  $N_0$  be the normal closure of  $\{y, t\}$  in  $Q_+$ . Then  $Q_+/N_0 = \langle x \rangle$  is infinite cyclic and the extension splits, so  $Q_+ = N_0 \rtimes \langle x \rangle$ . Rewriting the presentation of  $Q_+$  over the transversal  $\{x^n\}$  by the Reidemeister–Schreier method [LS, Chapter II] presents  $N_0$  on the generators  $y_n = x^n y x^{-n}$  and  $t_m = x^m t x^{-m}$ ,

$n, m \in \mathbb{Z}$ , with relators the conjugates  $x^m[t, y_n]x^{-m} = [t_m, y_{n+m}]$ , that is, all commutators  $[t_m, y_n]$ . Hence  $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$ . Now  $j(y_n) = (x_1^n y x_1^{-n}, 1)$  and  $j(t_m) = (1, x_2^m t x_2^{-m})$ , and the elements  $x_1^n y x_1^{-n}$  freely generate the normal closure of  $y$  in  $F(x_1, y)$ , and likewise for  $t$ ; so  $j$  restricted to  $N_0$  is an isomorphism onto the product of these two normal closures. Finally  $j(x) = (x_1, x_2)$  has infinite order and  $j(x)^k \notin j(N_0)$  for  $k \neq 0$ , because, for  $h \in N_0$ , the exponent sum of  $x_1$  in  $j(x^k h)$  is  $k$ . Hence  $j$  is injective on  $N_0 \rtimes \langle x \rangle$ . Free groups are residually finite and residual finiteness passes to direct products and subgroups, so  $Q_+$  is residually finite.  $\square$

For  $e \in \mathbb{N}$  let  $Q_e = B(C_e) = F/N_e$  with  $C_e$  the group of Lemma 5.2, and write  $q_e : F \rightarrow Q_e$ . By Lemmas 5.2 and 5.3, a recursive presentation of  $Q_e$  on  $x, y, t$  is computable from  $e$ ;  $N_e \leq N_+$ ; if  $e \in \text{INF}$  then  $Q_e = Q_+$ ,  $N_e = N_+$  and  $q_e = q_+$ ; and if  $e \in \text{FIN}$  then  $H \leq C_e \leq Q_e$ .

### Detecting $N_e$ inside a finitely presented group.

**Lemma 5.4** (Higman embedding and the Mihailova subgroup). *There is an algorithm that computes from  $e$  a finite presentation  $\langle X_e \mid R_e \rangle$  of a group  $\Lambda_e$ , words  $w_x, w_y, w_t \in F(X_e)$ , and a finite generating set of a subgroup  $M_e \leq F(X_e) \times F(X_e)$ , such that  $x \mapsto w_x, y \mapsto w_y, t \mapsto w_t$  induces an embedding  $Q_e \rightarrow \Lambda_e$ , and  $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$ . Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \quad i_e(f) = (f, w_f, 1), \quad L_e^0 = F \times M_e,$$

where  $w_f$  is the image of  $f$  under  $x \mapsto w_x, y \mapsto w_y, t \mapsto w_t$ , the map  $i_e : F \rightarrow K_e^0$  is injective and  $i_e(f) \in L_e^0$  if and only if  $f \in N_e$ .

*Proof.* Higman's embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of  $Q_e$ . Let  $M_e$  be the subgroup of  $F(X_e) \times F(X_e)$  generated by the pairs  $(z, z)$ ,  $z \in X_e$ , and  $(r, 1)$ ,  $r \in R_e$ ; then  $M_e = \{(u, v) : u = v \text{ in } \Lambda_e\}$  by Mihailova's lemma [Mih]. The first coordinate makes  $i_e$  injective, and  $(f, w_f, 1) \in F \times M_e$  if and only if  $w_f = 1$  in  $\Lambda_e$ , if and only if  $q_e(f) = 1$  because the embedding is injective, if and only if  $f \in N_e$ .  $\square$

**The two HNN extensions.** Let  $P = F(x_1, y) \times F(x_2, t)$  and  $j : Q_+ \rightarrow P$  be as in Lemma 5.3, and put

$$K^g = F \times P, \quad L^g = \{(f, jq_+(f)) : f \in F\},$$

and define

$$K_e = K_e^0 \times K^g, \quad L_e = L_e^0 \times L^g, \quad i : F \rightarrow K_e, \quad i(f) = (i_e(f), (f, 1)).$$

Let

$$(3) \quad \Gamma_e = \langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle, \quad S_e = \langle i(F), v i(F) v^{-1} \rangle \leq \Gamma_e.$$

**Lemma 5.5** (the first HNN extension).

- (1)  $K_e$  is a direct product of free groups of finite rank, hence finitely presented and residually finite;  $L_e$  is finitely generated;  $i$  is injective; and  $i(F) \cap L_e = i(N_e)$ .
- (2)  $\Gamma_e$  is finitely presented, and a finite presentation is computable from  $e$ . The natural map from the amalgamated free product  $F *_{N_e} F$ , in which  $n \in N_e$  in the first copy is identified with  $n$  in the second copy, onto  $S_e$ , sending the first copy to  $i(F)$  and the second to  $v i(F) v^{-1}$ , is an isomorphism.
- (3) There is a homomorphism  $\tau_e: S_e \rightarrow Q_e$  with  $\tau_e(i(f)) = q_e(f)$  and  $\tau_e(v i(f) v^{-1}) = 1$  for  $f \in F$ .

*Proof.* (1)  $K_e = F \times F(X_e) \times F(X_e) \times F \times P$  is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup  $L_e$  is generated by the basis of the first factor, the generators of  $M_e$  from Lemma 5.4, and the elements  $(a, jq_+(a))$  for  $a \in \{x, y, t\}$ , so it is finitely generated. Injectivity of  $i$  is clear from the first coordinate. An element  $i(f)$  lies in  $L_e$  if and only if  $i_e(f) \in L_e^0$  and  $(f, 1) \in L_e^g$ ; the first holds if and only if  $f \in N_e$  by Lemma 5.4, and the second if and only if  $jq_+(f) = 1$ , that is,  $f \in N_+$ . Since  $N_e \leq N_+$ , the intersection is  $i(N_e)$ .

(2) The presentation (3) is finite once  $[v, \ell] = 1$  is imposed only for the finitely many generators  $\ell$  of  $L_e$ , and it is computable from  $e$  because  $K_e$ ,  $L_e$  and  $i$  are. It is an HNN extension of  $K_e$  with both edge groups equal to  $L_e$  and the identity as edge isomorphism. The map from  $F *_{N_e} F$  to  $S_e$  is well defined because  $i(n) \in L_e$  commutes with  $v$  for  $n \in N_e$ . A reduced word of the amalgamated free product alternates between elements of the two copies of  $F$  lying outside  $N_e$ ; its image is a word  $i(f_0) v i(f_1) v^{-1} i(f_2) \cdots$  in which every letter between consecutive occurrences of  $v^{\pm 1}$  is some  $i(f_k)$  with  $f_k \notin N_e$ , hence  $i(f_k) \notin L_e$  by (1). Such a word contains no pinch  $v^{\pm 1} \ell v^{\mp 1}$  with  $\ell \in L_e$ , so it is nontrivial in  $\Gamma_e$  by Britton's lemma.

(3) Define  $\tau_e$  on  $F *_{N_e} F$  by  $q_e$  on the first copy and by the trivial map on the second; both kill  $N_e$ , so the definition is consistent on the amalgamated subgroup, and (2) identifies  $F *_{N_e} F$  with  $S_e$ .  $\square$

The second HNN extension is

$$(4) \quad R_e = \langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \tau_e(s)) \ (s \in S_e) \rangle,$$

whose edge maps  $s \mapsto (s, 1)$  and  $s \mapsto (s, \tau_e(s))$  are injective homomorphisms of  $S_e$  into  $\Gamma_e \times Q_e$ ; so  $\Gamma_e \times Q_e$  embeds in  $R_e$  by Britton's lemma. The presentation (4) lists the recursively enumerable relators of  $Q_e$ . Its finite replacement is

$$(5) \quad \begin{aligned} \widehat{R}_e = \langle \Gamma_e \times F, u \mid & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(v i(a) v^{-1}, 1)u^{-1} = (v i(a) v^{-1}, 1) \ (a \in \{x, y, t\}) \rangle, \end{aligned}$$

where  $\Gamma_e \times F$  is equipped with the finite presentation of Lemma 5.5(2), the basis  $x, y, t$  of  $F$ , and the commutation relators between the two factors.

**Lemma 5.6** (the finite presentation). *The groups  $\widehat{R}_e$  and  $R_e$  are isomorphic by the map that is the identity on  $\Gamma_e$  and on  $u$  and sends  $F$  onto  $Q_e$  by  $q_e$ . A code of the finite presentation (5) is computable from  $e$ .*

*Proof.* Both maps  $f \mapsto u(i(f), 1)u^{-1}$  and  $f \mapsto (i(f), f)$  are homomorphisms  $F \rightarrow \widehat{R}_e$ , and they agree on the basis, so  $u(i(f), 1)u^{-1} = (i(f), f)$  for all  $f \in F$ ; likewise  $u(vi(f)v^{-1}, 1)u^{-1} = (vi(f)v^{-1}, 1)$  for all  $f$ . If  $n \in N_e$ , then  $i(n) \in L_e$  by Lemma 5.5(1), so  $vi(n)v^{-1} = i(n)$  in  $\Gamma_e$ , and comparing the two relations gives  $(i(n), n) = (i(n), 1)$ , that is,  $(1, n) = 1$  in  $\widehat{R}_e$ . Hence the factor  $F$  of  $\widehat{R}_e$  factors through  $Q_e = F/N_e$ , and the relations of (5) become  $u(s, 1)u^{-1} = (s, \tau_e(s))$  for  $s$  in the generating set  $\{i(a), vi(a)v^{-1} : a \in \{x, y, t\}\}$  of  $S_e$ ; both sides define homomorphisms on  $S_e$ , so the relations hold for all  $s \in S_e$ . This defines a homomorphism  $R_e \rightarrow \widehat{R}_e$ ; conversely, the relations of (5) hold in  $R_e$  after  $F \rightarrow Q_e$ , because  $\tau_e(i(a)) = q_e(a)$  and  $\tau_e(vi(a)v^{-1}) = 1$ . The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 5.4 and 5.5.  $\square$

**Lemma 5.7** (the case  $e \in \text{FIN}$ ). *If  $e \in \text{FIN}$ , then  $H$  embeds in  $\widehat{R}_e$ , and  $\widehat{R}_e$  is not MF.*

*Proof.* For  $e \in \text{FIN}$  we have  $H \leq C_e \leq Q_e \leq \Gamma_e \times Q_e \leq R_e \cong \widehat{R}_e$  by Lemmas 5.2, 5.3(1), and 5.6, and Britton's lemma for (4). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and  $H$  is not MF by Theorem B of the companion manuscript [NMF].  $\square$

## 6. THE CASE $e \in \text{INF}$

It remains to prove that  $\widehat{R}_e$  is MF for  $e \in \text{INF}$  and to deduce Theorem A. Its tracial MF realization lies in a norm reduced product of MF algebras. For  $C^*$ -algebras  $B_n$ ,  $n \geq 1$ , write  $\prod_n B_n / \bigoplus_n B_n$  for the quotient of the bounded sequences by the sequences with  $\|x_n\| \rightarrow 0$ ; its norm is  $\limsup_n \|x_n\|$ .

**Lemma 6.1** (reduced products of MF algebras). *If every  $B_n$  is MF, then every separable  $C^*$ -subalgebra of  $\prod_n B_n / \bigoplus_n B_n$  is MF. Moreover,  $M_k(B)$  is MF whenever  $B$  is.*

*Proof.* The second assertion holds because  $M_k$  of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by  $k$ . For the first, let  $A$  be a separable  $C^*$ -subalgebra, and choose a sequence  $(a_i)_{i \geq 1}$  dense in  $A$ , with lifts  $(a_i^{(n)})_n$ . Choose for each  $n$  an injective  $*$ -homomorphism  $\eta_n$  of  $B_n$  into a norm matrix corona, and lifts  $(a_i^{(n,k)})_k$  of  $\eta_n(a_i^{(n)})$  with  $\sup_k \|a_i^{(n,k)}\| \leq \|a_i^{(n)}\| + 1$ . Enumerate the noncommutative  $*$ -polynomials with coefficients in  $\mathbb{Q}(i)$  as  $p_1, p_2, \dots$ , and let  $\mathcal{P}_n$  be the set of those among  $p_1, \dots, p_n$  that involve only  $a_1, \dots, a_n$ , choosing  $p_1 = 0$  so that  $\mathcal{P}_n$  is

nonempty. For  $p \in \mathcal{P}_n$  the number  $\|p(a^{(n)})\|$  equals  $\limsup_k \|p(a^{(n,k)})\|$ , since  $\eta_n$  is isometric. Hence there is  $K_n$  such that  $\|p(a^{(n,k)})\| \leq \|p(a^{(n)})\| + 1/n$  for all  $k \geq K_n$  and  $p \in \mathcal{P}_n$ , and for each  $p \in \mathcal{P}_n$  there is  $k_p \geq K_n$  with  $\|p(a^{(n,k_p)})\| \geq \|p(a^{(n)})\| - 1/n$ . Put

$$b_i^{(n)} = \bigoplus_{p \in \mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For  $p \in \mathcal{P}_n$  the norm of  $p(b^{(n)})$  is the maximum of  $\|p(a^{(n,k_{p'})})\|$  over  $p' \in \mathcal{P}_n$ , which lies between  $\|p(a^{(n)})\| - 1/n$  and  $\|p(a^{(n)})\| + 1/n$ . Every polynomial belongs to  $\mathcal{P}_n$  for all large  $n$ , so  $\limsup_n \|p(b^{(n)})\| = \limsup_n \|p(a^{(n)})\| = \|p(a)\|$  for every  $p$ . Therefore  $a_i \mapsto [(b_i^{(n)})_n]$  is well defined and isometric on the  $*$ -algebra over  $\mathbb{Q}(i)$  generated by the  $a_i$ , which is dense in  $A$ , and it extends to an isometric  $*$ -homomorphism of  $A$  into a norm matrix corona.  $\square$

**Lemma 6.2** (tensor synchronization). *Let  $\Gamma$  be a countable group with a tracial MF realization  $(A_1, \rho_1, \tau_1)$ , let  $Q$  be a countable group with homomorphisms  $\beta_n: Q \rightarrow B_n$  to finite groups such that every  $q \neq 1$  satisfies  $\beta_n(q) \neq 1$  for all large  $n$ , let  $S \leq \Gamma$ , and let  $\tau: S \rightarrow Q$  be a homomorphism. Suppose that for every  $n$  there is a homomorphism  $\lambda_n: \Gamma \rightarrow G_n$  to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

*Then there are a separable unital MF  $C^*$ -algebra  $A$ , a homomorphism  $V: \Gamma \times Q \rightarrow \mathcal{U}(A)$ , a tracial state  $T$  on  $A$  with  $T(V(g, q)) = 0$  for  $(g, q) \neq (1, 1)$ , and a unitary  $W \in A$  with  $WV(s, 1)W^* = V(s, \tau(s))$  for all  $s \in S$ . Consequently*

$$\langle \Gamma \times Q, u \mid u(s, 1)u^{-1} = (s, \tau(s)) \ (s \in S) \rangle$$

*admits a tracial MF realization, and in particular is MF.*

*Proof.* Let  $E_n$  be the image of  $(\lambda_n, \beta_n): \Gamma \times Q \rightarrow G_n \times B_n$ , a finite group of order  $k_n$ , and let  $L_n$  be its left regular representation on  $\ell^2(E_n)$ . Put  $B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$ , which is MF by Lemma 6.1, let  $E = \prod_n B'_n / \bigoplus_n B'_n$ , and define

$$V(g, q) = [(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q)))_n], \quad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \text{tr}_{k_n})(x_n),$$

for a free ultrafilter  $\omega$ . Each coordinate of  $V$  is a homomorphism into a unitary group, so  $V$  is a homomorphism, and  $T$  is a tracial state on  $E$  because  $|\tau_1 \otimes \text{tr}_{k_n}(x_n)| \leq \|x_n\|$ . If  $g \neq 1$  then  $\tau_1(\rho_1(g)) = 0$ , so  $T(V(g, q)) = 0$ ; if  $g = 1$  and  $q \neq 1$  then  $\beta_n(q) \neq 1$  for all large  $n$ , the regular representation of a finite group has trace zero off the identity, and again  $T(V(1, q)) = 0$ .

The two homomorphisms  $s \mapsto (\lambda_n(s), 1)$  and  $s \mapsto (\lambda_n(s), \beta_n \tau(s))$  from  $S$  to  $E_n$  have the same kernel, namely  $\ker(\lambda_n|_S)$ , by the hypothesis. Their images are therefore isomorphic subgroups of  $E_n$  of the same order, isomorphic via  $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n \tau(s))$ . The restriction of  $L_n$  to a subgroup of  $E_n$  is

unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary  $W_n \in M_{k_n}(\mathbb{C})$  with  $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n \tau(s))$  for all  $s \in S$ . Put  $W = [(1 \otimes W_n)_n] \in E$ ; then  $WV(s, 1)W^* = V(s, \tau(s))$  for  $s \in S$ . Let  $A$  be the  $C^*$ -algebra generated by  $V(\Gamma \times Q)$  and  $W$ ; it is separable and unital, and MF by Lemma 6.1. The triple  $(A, V, T|_A)$  is a tracial MF realization of  $\Gamma \times Q$ . For the last assertion, apply Theorem 4.2 to this realization. Let  $\iota$  be an injective  $*$ -homomorphism of  $A$  into a norm matrix corona and put  $p = \iota(1)$ . The element  $\tilde{W} = \iota(W) + (1 - p)$  is a unitary of the full corona and implements the required covariance on  $\iota(A)$ , so it is the required conjugator.  $\square$

**Lemma 6.3** (the case  $e \in \text{INF}$ ). *If  $e \in \text{INF}$ , then  $\hat{R}_e$  admits a tracial MF realization; in particular, it is MF.*

*Proof.* Let  $e \in \text{INF}$ , so that  $Q_e = Q_+$ ,  $N_e = N_+$ ,  $q_e = q_+$ , and  $\tau_e(i(f)) = q_+(f)$ ,  $\tau_e(v i(f) v^{-1}) = 1$ . By Lemma 5.6 it suffices to show that  $R_e$  admits a tracial MF realization, and by Lemma 6.2 applied to  $\Gamma = \Gamma_e$ ,  $Q = Q_+$ ,  $S = S_e$  and  $\tau = \tau_e$ , it suffices to produce the data of that lemma.

The group  $K_e$  is residually finite by Lemma 5.5(1), hence admits a tracial MF realization by Lemma 4.1, and  $\Gamma_e$  is the central HNN extension of  $K_e$  over  $L_e$ , hence admits one by Corollary 4.3.

The group  $P = F(x_1, y) \times F(x_2, t)$  is residually finite; choose finite-index normal subgroups  $P_1 \geq P_2 \geq \dots$  of  $P$  with trivial intersection, let  $r_n: P \rightarrow C_n = P/P_n$ , and put  $\beta_n = r_n \circ j: Q_+ \rightarrow C_n$  with  $j$  the embedding of Lemma 5.3(3). Since  $j$  is injective and the  $P_n$  decrease to the trivial group, every  $h \neq 1$  has  $\beta_n(h) \neq 1$  for all large  $n$ .

Let  $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$ , where  $\sigma$  of order two exchanges the coordinates. Define  $\lambda_n$  on  $K_e = K_e^0 \times (F \times P)$  by killing  $K_e^0$  and sending  $(f, p)$  to  $(r_n j q_+(f), r_n(p))$ , and put  $\lambda_n(v) = \sigma$ . This respects the relations of (3): the factor  $L_e^0$  of  $L_e$  lies in  $K_e^0$  and is killed, and an element  $(f, j q_+(f))$  of  $L^g$  is sent to  $(r_n j q_+(f), r_n j q_+(f))$ , which lies on the diagonal and commutes with  $\sigma$ . Hence  $\lambda_n: \Gamma_e \rightarrow G_n$  is a homomorphism, and on the generators of  $S_e$ ,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \quad \lambda_n(v i(f) v^{-1}) = (1, r_n j q_+(f)).$$

Let  $\pi_0, \pi_1: S_e \rightarrow Q_+$  be the homomorphisms that are  $q_+$  on one copy of  $F$  in  $S_e \cong F *_{N_+} F$  and trivial on the other, as in Lemma 5.5(3); thus  $\pi_0 = \tau_e$ . The displayed formulas show that  $\lambda_n|_{S_e} = (r_n j \pi_0, r_n j \pi_1)$ , both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \tau_e),$$

which is the hypothesis of Lemma 6.2.  $\square$

*Proof of Theorem A.* By Lemma 5.6 the map  $e \mapsto \hat{R}_e$  is computable, by Lemma 6.3 the presented group is MF for  $e \in \text{INF}$ , and by Lemma 5.7 it is not MF for  $e \in \text{FIN}$ . Hence  $\text{INF}$  reduces to  $\text{MF}_{\text{fp}}$  and  $\text{FIN}$  reduces to  $\text{NONMF}_{\text{fp}}$  under computable many-one reductions. Since  $\text{INF}$  is  $\Pi_2^0$ -complete and  $\text{FIN}$



is  $\Sigma_2^0$ -complete, and since  $\text{MF}_{\text{fp}} \in \Pi_2^0$  and  $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$  by Proposition 3.2, the two sets are complete at their levels. A  $\Pi_2^0$ -complete set is not  $\Sigma_2^0$  and a  $\Sigma_2^0$ -complete set is not  $\Pi_2^0$ , so neither set is recursively enumerable, and in particular neither is decidable.  $\square$

*Remark 6.4.* The construction uses only these properties of  $H$ : it is finitely generated, recursively presented, and not MF, while subgroups of MF groups are MF. It therefore applies to any group property that is closed under subgroups, given a finitely generated recursively presented group without the property and an analogue of Lemma 6.2 for the twisted HNN extensions; the open problems below record which input is missing for soficity and for hyperlinearity.

The homomorphism  $V$  of Lemma 6.2 does not extend faithfully to  $R_e$  in the corona: the finite maps  $\lambda_n$  are trivial on the Mihailova factor, so  $W$  commutes with  $V(\gamma, 1)$  for every  $\gamma \in K_e^0$ , although  $u$  does not commute with  $(\gamma, 1)$  in  $R_e$ . Injectivity is obtained in Theorem 4.2 from the universal algebra.

**Open problems.** We do not know whether the models of  $\widehat{R}_e$  can be chosen to converge to  $C_r^*(\widehat{R}_e)$  in operator norm, so that the reduced  $C^*$ -algebra itself is MF; the argument produces only a tracial state on an abstract MF algebra. We do not know whether  $\widehat{R}_e$  is hyperlinear for  $e \in \text{INF}$ ; placing hyperlinearity recognition at the same level would also require a finitely presented non-hyperlinear group, and none is known. We do not know whether an analogous family of finitely presented groups exists for soficity; such a construction would require a permutation-model replacement for Lemma 6.2. Finally, we do not know the complexity of MF recognition under the strong convergence convention of [GKM, Sch]: the norms  $\|\lambda(w)\|$  in the left regular representation are not in general computable from a finite presentation, so Proposition 3.2 does not apply.

#### ACKNOWLEDGMENTS

The author thanks Francesco Fournier-Facio for early advice and detailed comments on earlier drafts, and Alon Dogon for organizing the ensuing discussion and for comments on the surrounding literature. The author also thanks Tatiana Shulman for clarifying the terminology and historical context of the MF question, and Caleb Eckhardt for discussions of MF representations and matrix norms.

#### USE OF AI AND FORMAL METHODS

Claude Fable 5 and GPT-5.6 Sol assisted with mathematical exploration, proof development, literature searches, Lean formalization, drafting, and editing. Lean 4 source is available at <https://github.com/SauersML/group-approximation>.

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